



C23-EC-401

23430

BOARD DIPLOMA EXAMINATION, (C-23)

MARCH/APRIL—2025

DECE – FOURTH SEMESTER EXAMINATION

ELECTRONIC CIRCUITS—II

Time : 3 hours]

[Total Marks : 80

PART—A

3×10=30

- Instructions :**
- (1) Answer **all** questions.
 - (2) Each question carries **three** marks.
 - (3) Answers should be brief and straight to the point and shall not exceed five simple sentences.

- 1. List different types of linear and non-linear wave shaping circuits.
- 2. List any six merits of SMT technology.
- 3. State the concept of virtual ground.
- 4. List any six characteristics of ideal operational amplifier.
- 5. Define sweep voltage and state its use as time-base.
- 6. List any three applications of voltage to current converter.
- 7. Define capture range of PLL.
- 8. Draw the pin diagram of 555 IC.
- 9. Define the terms (a) accuracy, (b) settling time of D/A converter.
- 10. List IC numbers of any three ADCs.

/23430

1

[Contd...

www.manareresults.co.in

www.manareresults.co.in

PART—B

10×5=50

- Instructions :** (1) Answer *any five* questions.
(2) Each question carries **ten** marks.
(3) Answers should be comprehensive and the criteria for valuation is the content but not the length of the answer.

- 11.** Explain RC differentiator circuit with waveforms. 10
- 12.** Explain the working of positive and negative unbiased diode clipper circuits with waveforms. 10
- 13.** (a) Draw the pin diagram of IC 741. 3
(b) Explain the function of operational amplifier as inverter amplifier with a circuit diagram and derive the expression for voltage gain of inverting amplifier. 7
- 14.** Explain the working of Op-amp based Wien-bridge oscillator circuit and state the conditions required for stable operations. 10
- 15.** Explain the working of Op-amp based Bootstrap sweep circuit. 10
- 16.** Draw and explain the working of astable multi-vibrator using 555 IC. 10
- 17.** Draw the internal block diagram of PLL-LM 565 and explain its working. 10
- 18.** Explain D/A conversion using binary weighted resistors. 10

★ ★ ★